# IRVAN TRIANTO

GIS & SPATIAL DATA ANALYST | MAPPING & NAVIGATION | ENVIRONMENTAL & MARINE GIS
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## PROFESSIONAL SUMMARY

Geospatial professional with hands-on experience in GIS, spatial data management, digital mapping, hydrographic surveying, coordinate validation, and geospatial project support across infrastructure, marine/coastal, conservation, and academic environments. Experienced with QGIS, ArcGIS/ArcGIS Pro, GNSS/GPS positioning, spatial QA/QC, Excel/CSV, GeoJSON, SHP, KML, and PostgreSQL/PostGIS. Developing Python/Pandas capabilities for reproducible data cleaning and validation workflows, with working knowledge of Leaflet, WebGIS and Google Earth Engine (GEE) for satellite based vegetation monitoring. Strong field-to-office workflow experience and cross-functional communication.

## CORE COMPETENCIES

- **GIS & Mapping:** QGIS, ArcGIS, ArcGIS Pro, Google Earth Engine (GEE), Global Mapper, ENVI, thematic mapping, spatial analysis, map production
- **Spatial Data:** SHP, GeoJSON, KML, Excel/CSV, coordinate validation, data structuring, QA/QC
- **Positioning & Survey:** GNSS/GPS, hydrographic survey, ADCP, Current Meter, field coordinate verification
- **Data & Database:** Excel data cleaning, PostgreSQL/PostGIS (basic), SQL fundamentals, Python/Pandas (developing)
- **WebGIS:** Leaflet, GeoJSON, HTML/CSS/JavaScript fundamentals, WebGIS visualization
- **Domain:** Marine/coastal/fisheries GIS, environmental mapping, conservation, infrastructure projects

## PROFESSIONAL EXPERIENCE

**PT Wahyudi Andy Laksito Setiarso** | *Kalimantan Timur, Indonesia.* *Aug 2026 – Present*

*GIS Operator — Zonasi Nilai Tanah Kabupaten Bulungan, Kalimantan Utara*

- Coordinating the surveyor team and validating land sample points collected in the field.
- Performing quality control by calculating the standard deviation of land prices to ensure data accuracy.
- Managing and authorizing the final field data through the SIPENTA information system.
- Processing spatial data using ArcGIS Pro to produce the final Land Value Zone Maps.
- Developing and managing a web-based spatial dashboard (https://itrianto.github.io/znt-bulungan-2026/) for data visualization and monitoring.

**PT Gema Sarana Teknik** |*Batam, Indonesia.* *Aug 2025 – Aug 2026*

*GIS Surveyor — Navigation, Positioning & Geospatial Data Processing*

- Conducted geospatial and hydrographic data collection, positioning, and digital mapping across survey projects, including a 50 days deployment supporting the Karangkates Floating Photovoltaic Solar Power Plant (100 MWac), a PLN infrastructure project.
- Processed satellite imagery and produced thematic spatial outputs, including nautical charts, mangrove distribution maps, and seagrass (lamun) mapping by integrating field observations with spatial datasets.
- Operated GNSS/GPS positioning systems and hydrographic survey instruments in offshore, coastal, and dynamic field environments; applied coordinate and projection-system knowledge.
- Performed spatial data validation and cross-verification between field observations, satellite imagery, and CAD/survey records to improve accuracy and consistency of geospatial deliverables.
- Coordinated with cross-functional field and technical teams under project specifications, supporting technical reporting and operational deliverables.

**PT Amati Karya Indonesia** | *Jakarta, Indonesia.* *Feb 2024 – Jun 2024*

*Project Coordinator — Coral Reef Conservation Team*

- Processed and interpreted satellite imagery for environmental monitoring, spatial analysis, and coastal ecosystem mapping supporting conservation planning.

- Compiled, analyzed, and visualized spatial data into thematic maps for field-based project decision making.
- Coordinated multidisciplinary researchers, field staff, and community stakeholders across GIS-based environmental assessment activities.
- Presented spatial analysis results and project outcomes at an international forum.

**Universitas Negeri Semarang** | *Semarang, Indonesia.* *Sep 2024 – Sep 2025*

*Lab Supervisor*

- Supported Remote Sensing practicum sessions covering satellite imagery processing, image interpretation, GIS analysis, data preparation, and software troubleshooting.
- Verified and maintained 100+ academic records through systematic quality-control procedures, achieving 100% data compliance.
- Oversaw laboratory operations and adherence to procedures and safety standards during practical sessions.

**Measuring Current Project** | *Batam, Indonesia.* *Feb 2026*

*Hydrology Surveyor — Freelance Project*

- Conducted offshore environmental field surveys using ADCP and Current Meter instruments under dynamic sea conditions.
- Performed QA and validation of hydrodynamic observations to support environmental and engineering risk assessment.

## ADDITIONAL PROJECT EXPERIENCE

**Community Service Program** | *Karanganyar, Indonesia.* *Dec 2024 – Jan 2025*

*Project Coordinator*

- Led 15 community development projects, integrating social and environmental safety considerations into field planning.
- Coordinated technical activities and supervised field implementation across multidisciplinary community programs.

**Study of Geosphere Phenomena 2** | *Karimunjawa Island, Indonesia.* *Feb – Jun 2023*

*Project Coordinator*

- Led a nearshore environmental survey assessing coastal abrasion and shoreline change, producing spatial and field-based evidence for environmental assessment.
- Implemented field safety controls including life-jacket protocols and weather/wave checks before coastal survey activities.

## EDUCATION

**Universitas Negeri Semarang — Bachelor's Degree in Geography** | Semarang, Indonesia. 2021 – 2025

GPA: 3.79/4.00 | Top 10 Finalist, Faculty Ambassador (FISIP), 2022.

## SELECTED PUBLICATIONS

- Connectivity Index Analysis of Religious Tourism Sites in Semarang City to Support Tourist Visits. GeoImage Journal, Vol. 15 No. 1 (2026). DOI: 10.15294/geoimage.v15i1.40537
- Trichoderma sp. Cultivation Training as a Natural Fungicide, Green Lawu Forest Farmer Group, Karanganyar. ABDIMAS: Jurnal Pengabdian Masyarakat, Vol. 29 No. 1 (2025). DOI: 10.15294/q7khw835
- The Effect of Land Cover on the Urban Index of Semarang City. Proceedings of the FISIP UNNES National Seminar (2024).

## CERTIFICATIONS

- Ahli K3 Umum (AK3U) Training, 2026 — PT Sarana Inspirasi Maju Bersaudara (Professional Project Institute), Batam; official Kemnaker certification in issuance process.
- International Certificate in Sustainable Marine Ecosystems, 2024 — Ghent University & ASEAN South Asia Platform.
- TOPEL — Score 443, 2025.
- Internship Certification — IS-CORE Wallacea Project, 2024 — PT Amati Karya Indonesia in collaboration with ASEAN South Asia Platform.